

MATPREDICT · MATING-TYPE REPORT

basidio_serpula_HD_PR

taxid 936435

Run by MATPredict 0.6.1 · database 2b90ce832408 · NCBI E-utilities · rendered 2026-10-08 04:24 UTC by MATPredict 0.6.1

Result

MATING-TYPE LOCI

undetermined HD **high confidence****undetermined** PR **medium confidence****undetermined** PR **medium confidence**

- **undetermined** (HD family): complete locus, high confidence, 32.1 kb on GL945494.1. Gene order: MIP1 · **HD1** · **HD2** · beta_fg.
- **undetermined** (PR family): MAT gene without flanking genes, medium confidence, 30.5 kb on GL945477.1. Gene order: **fungal_mating_type_pheromone** · **pheromone_receptor** · **caax_precursor**.
- **undetermined** (PR family): MAT gene without flanking genes, medium confidence, 49.2 kb on GL945501.1. Gene order: **fungal_mating_type_pheromone** · **caax_precursor** · **pheromone_B43** · **pheromone_receptor**.

Called loci

3

undetermined

HD family · Basidiomycota

high confidence

160 kb

170 kb

180 kb

190 kb

called locus

69.1 kb more to contig end →

MIP1

HD1

HD2

beta_fg

Gene order on GL945494.1.

core MAT gene

conserved flank

tools disagree

alternate model

Check:

Idiomorph undetermined: the evidence for two idiomorphs is tied or too close to call.

LOCATION

LOCUS

GENES MODELLED

GL945494.1:160,053–192,142 ·
32.1 kb

Complete locus: MAT gene(s)
with their flanking genes.

4 of 4 gene hits

NEAREST CONTIG END

REGION (SAMTOOLS)

70.7 kb

GL945494.1:160053-192142

GENE EVIDENCE

Gene	Role	Model	Identity %	E-value	Position (strand)	Exons	Reference record
conserved flank MIP1	conserved flank	models differ exonerate miniprot: 160,053–162,538, 68.1%	69.3	< 1e-180 419 bits	160,053–162,535 +	5	192523_h97_HD_A1
core MAT HD1	core MAT	models differ exonerate miniprot: 162,954–165,228, 39.0%	49.9	1.2e-99 329 bits	163,632–164,804 -	3	90004_unknown_HD_A1
core MAT HD2	core MAT	models differ exonerate miniprot: 165,575–167,421, 39.9%	44.5	9.2e-73 219 bits	166,414–167,376 +	2	90004_unknown_HD_A1
conserved flank beta_fg	conserved flank	models differ exonerate miniprot: 168,402–169,313, 48.2%	68.3	3.5e-12 66 bits	168,402–168,638 -	1	192523_h97_HD_A1

undetermined

PR family · Basidiomycota

medium confidence

2,815 kb

2,820 kb

2,825 kb

2,830 kb

2,835 kb

2,840 kb

2,845 kb

called locus

278.8 kb more to contig end →

pheromone receptor

caax_precursor

al_mating_type_pheromone

Gene order on GL945477.1.

core MAT gene

tools disagree

alternate model

search hit only

Check:

The pheromone-precursor gene was found only by a sequence-motif scan (CAAX), with no reference match. Treat this call as unconfirmed.

Check:

Idiomorph undetermined: the evidence for two idiomorphs is tied or too close to call.

Part of pheromone-receptor array Basidiomycota:PR:GL945477.1:2839835-2841105 (1 locus, unsupported). Arrays hold mating and non-mating receptors, so membership is not evidence of mating function.

LOCATION

GL945477.1:2,815,887–2,846,352 · 30.5 kb

LOCUS

A MAT gene with no flanking gene beside it; not a full locus.

GENES MODELLED

2 of 3 gene hits

NEAREST CONTIG END

280.3 kb

REGION (SAMTOOLS)

GL945477.1:2815887-2846352

GENE EVIDENCE

Gene	Role	Model	Identity %	E-value	Position (strand)	Exons
core MAT fungal_mating_type_pheromone	core MAT	hit only tblastn	34.2 cov 55.0%	3.0e+00 26 bits	2,815,887–2,816,000 -	-
core MAT pheromone_receptor	core MAT	models differ exonerate miniprot: 2,813,932–2,841,108, 46.6%	54.9	5.9e-74 175 bits	2,839,871–2,841,102 -	6
core MAT caax_precursor	core MAT	caax orf motif scan (CAAX) motif CVIG, 40 aa ORF	0.0	-	2,846,230–2,846,352 +	-
pheromone_B43	expected, not found					
pheromone_B44	expected, not found					

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undetermined PR family · Basidiomycota

medium confidence

20 kb30 kb40 kb50 kb60 kb

← 14.9 kb more to contig end

called locus

pheromone_receptor

al_mating_type_pheromone

caax_precursor

pheromone_B43

Gene order on GL945501.1.

core MAT gene

tools disagree

alternate model

search hit only

Check:

The pheromone-precursor gene was found only by a sequence-motif scan (CAAX), with no reference match. Treat this call as unconfirmed.

Check:

Idiomorph undetermined: the evidence for two idiomorphs is tied or too close to call.

Part of pheromone-receptor array Basidiomycota:PR:GL945501.1:42260-66616 (5 loci, supported). Arrays hold mating and non-mating receptors, so membership is not evidence of mating function.

LOCATION

GL945501.1:17,376–66,616 · 49.2 kb

LOCUS

A MAT gene with no flanking gene beside it; not a full locus.

GENES MODELLED

2 of 4 gene hits

NEAREST CONTIG END

17.4 kb

REGION (SAMTOOLS)

GL945501.1:17376-66616

GENE EVIDENCE

Gene	Role	Model	Identity %	E-value	Position (strand)	Exons	Reference
fungus_mating_type_pheromone	core MAT	hit only tblastn	40.5 cov 100.0%	3.0e-03 33 bits	17,376–17,501 -	-	5627_90
caax_precursor	core MAT	caax orf motif scan (CAAX) motif CVIA, 43 aa ORF	0.0	-	46,358–46,489 +	-	caax_s
pheromone_B43	core MAT	hit only tblastn	38.3 cov 92.0%	8.0e-05 38 bits	53,498–53,650 -	-	5346_a okayama
pheromone_receptor	core MAT	models differ exonerate miniprot: 65,087–66,604, 52.3%	65.4	2.2e-115 308 bits	65,087–66,244 +	5	984962 B1
pheromone_B44	expected, not found						

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What was searched

ROUTING

By taxonomy. The genome's taxonomy falls inside a curated family's scope; only those families were searched.

FAMILIES SEARCHED

Basidiomycota:HD,
Basidiomycota:PR

GENETIC CODE

1

TAXONOMY

NCBI E-utilities

PHEROMONE-RECEPTOR ARRAYS

Receptor arrays occur for both mating and non-mating STE3-like receptors (the study found arrays holding paralogs beside the mating receptors in *Coprinopsis cinerea* and *Schizophyllum commune*), so array membership does not establish that a locus is a mating receptor.

receptor_array_support is a flag on the evidence behind an array; it never changes a call, its confidence or its verification label.

array	location	loci	calls	support	reasons
Basidiomycota:PR:GL945474.1:4713912-4714208	GL945474.1: 4,713,912–4,714,208	1	0	unsupported	a single receptor locus; no pheromone-precursor homology hit; 0 strict-CAAX pheromone ORFs
Basidiomycota:PR:GL945474.1:5389600-5394375	GL945474.1: 5,389,600–5,394,375	2	0	supported	array of 2 or more receptor loci
Basidiomycota:PR:GL945475.1:35023-36319	GL945475.1: 35,023–36,319	1	0	unsupported	a single receptor locus; no pheromone-precursor homology hit; 0 strict-CAAX pheromone ORFs
Basidiomycota:PR:GL945475.1:535802-537196	GL945475.1: 535,802–537,196	1	0	unsupported	a single receptor locus; no pheromone-precursor homology hit; 0 strict-CAAX pheromone ORFs
Basidiomycota:PR:GL945475.1:1033223-1034555	GL945475.1: 1,033,223–1,034,555	1	0	unsupported	a single receptor locus; no pheromone-precursor homology hit; 0 strict-CAAX pheromone ORFs

array	location	loci	calls	support	reasons
Basidiomycota:PR:GL945475.1:1368498-1369750	GL945475.1: 1,368,498–1,369,750	1	0	unsupported	a single receptor locus; no pheromone-precursor homology hit; 0 strict-CAAX pheromone ORFs
Basidiomycota:PR:GL945477.1:359515-360748	GL945477.1: 359,515–360,748	1	0	unsupported	a single receptor locus; no pheromone-precursor homology hit; 0 strict-CAAX pheromone ORFs
Basidiomycota:PR:GL945477.1:2839835-2841105	GL945477.1: 2,839,835–2,841,105	1	1	unsupported	a single receptor locus; no pheromone-precursor homology hit; 1 strict-CAAX pheromone ORF
Basidiomycota:PR:GL945477.1:3072958-3074228	GL945477.1: 3,072,958–3,074,228	1	0	unsupported	a single receptor locus; no pheromone-precursor homology hit; 0 strict-CAAX pheromone ORFs
Basidiomycota:PR:GL945478.1:522742-524014	GL945478.1: 522,742–524,014	1	0	unsupported	a single receptor locus; no pheromone-precursor homology hit; 0 strict-CAAX pheromone ORFs
Basidiomycota:PR:GL945478.1:1384609-1392726	GL945478.1: 1,384,609–1,392,726	3	0	supported	array of 2 or more receptor loci
Basidiomycota:PR:GL945481.1:438312-514117	GL945481.1: 438,312–514,117	5	0	supported	array of 2 or more receptor loci
Basidiomycota:PR:GL945481.1:744959-763356	GL945481.1: 744,959–763,356	2	0	supported	array of 2 or more receptor loci
Basidiomycota:PR:GL945485.1:1485124-1486281	GL945485.1: 1,485,124–1,486,281	1	0	unsupported	a single receptor locus; no pheromone-precursor homology hit; 0 strict-CAAX pheromone ORFs

array	location	loci	calls	support	reasons
Basidiomycota:PR:GL945487.1:506177-507452	GL945487.1: 506,177–507,452	1	0	unsupported	a single receptor locus; no pheromone-precursor homology hit; 0 strict-CAAX pheromone ORFs
Basidiomycota:PR:GL945488.1:432901-434194	GL945488.1: 432,901–434,194	1	0	unsupported	a single receptor locus; no pheromone-precursor homology hit; 0 strict-CAAX pheromone ORFs
Basidiomycota:PR:GL945488.1:786372-787667	GL945488.1: 786,372–787,667	1	0	unsupported	a single receptor locus; no pheromone-precursor homology hit; 1 strict-CAAX pheromone ORF
Basidiomycota:PR:GL945488.1:1048033-1049268	GL945488.1: 1,048,033–1,049,268	1	0	unsupported	a single receptor locus; no pheromone-precursor homology hit; 0 strict-CAAX pheromone ORFs
Basidiomycota:PR:GL945489.1:329536-330787	GL945489.1: 329,536–330,787	1	0	unsupported	a single receptor locus; no pheromone-precursor homology hit; 0 strict-CAAX pheromone ORFs
Basidiomycota:PR:GL945494.1:176372-177451	GL945494.1: 176,372–177,451	1	0	unsupported	a single receptor locus; no pheromone-precursor homology hit; 0 strict-CAAX pheromone ORFs
Basidiomycota:PR:GL945501.1:42260-66616	GL945501.1: 42,260–66,616	5	1	supported	array of 2 or more receptor loci; a pheromone-precursor homology hit; caax orfs>=2

array	location	loci	calls	support	reasons
Basidiomycota:PR:GL945525.1:32929-34269	GL945525.1: 32,929–34,269	1	0	unsupported	a single receptor locus; no pheromone-precursor homology hit; 0 strict-CAAX pheromone ORFs
Basidiomycota:PR:GL945543.1:3212-4569	GL945543.1: 3,212–4,569	1	0	unsupported	a single receptor locus; no pheromone-precursor homology hit; 0 strict-CAAX pheromone ORFs

Withheld candidate loci 38

Places with some MAT-gene evidence that did not pass a quality gate. Most genomes have several. They are listed so a known locus can be checked against them, not because they are likely calls.

Reasons: fewer than 2 genes could be modelled (36); too few of the family's genes found (2).

WITHHELD LOCI (38)

family	location	idiomorph	genes	modelled	reason
HD	GL945494.1: 112,924–113,127	undetermined	HD2	0	fewer than 2 genes could be modelled
HD	GL945481.1: 347,130–362,558	undetermined	HD2	0	fewer than 2 genes could be modelled
HD	GL945481.1: 1,068,836–1,068,997	undetermined	HD1, HD2	0	fewer than 2 genes could be modelled
HD	GL945485.1: 134,377–134,550	undetermined	HD1	0	fewer than 2 genes could be modelled
HD	GL945485.1: 792,088–792,321	undetermined	HD2	0	fewer than 2 genes could be modelled
HD	GL945485.1: 994,187–994,321	undetermined	HD2	0	fewer than 2 genes could be modelled
HD	GL945476.1: 337,894–338,034	undetermined	HD2	0	fewer than 2 genes could be modelled
HD	GL945476.1: 1,324,647–1,324,778	undetermined	HD2	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
HD	GL945484.1: 1,139,478–1,139,630	undetermined	HD1	0	fewer than 2 genes could be modelled
HD	GL945484.1: 1,208,343–1,208,690	undetermined	HD1	0	fewer than 2 genes could be modelled
HD	GL945484.1: 1,318,514–1,318,741	undetermined	HD2	0	fewer than 2 genes could be modelled
HD	GL945490.1: 295,907–296,134	undetermined	HD1	0	fewer than 2 genes could be modelled
HD	GL945496.1: 39,024–39,362	undetermined	HD1	0	fewer than 2 genes could be modelled
HD	GL945474.1: 455,847–456,113	undetermined	HD1	0	fewer than 2 genes could be modelled
HD	GL945474.1: 904,455–904,598	undetermined	HD2	0	fewer than 2 genes could be modelled
HD	GL945474.1: 2,863,441–2,863,566	undetermined	HD2	0	fewer than 2 genes could be modelled
HD	GL945474.1: 5,300,113–5,300,448	undetermined	HD1	0	fewer than 2 genes could be modelled
HD	GL945479.1: 721,111–721,299	undetermined	HD1	0	fewer than 2 genes could be modelled
HD	GL945479.1: 958,916–959,098	undetermined	HD2	0	fewer than 2 genes could be modelled
HD	GL945480.1: 1,159,799–1,159,993	undetermined	HD2	0	fewer than 2 genes could be modelled
HD	GL945480.1: 2,583,556–2,583,864	undetermined	HD1	0	fewer than 2 genes could be modelled
HD	GL945477.1: 911,465–911,890	undetermined	HD2	0	fewer than 2 genes could be modelled
HD	GL945477.1: 1,284,747–1,284,863	undetermined	HD1	0	fewer than 2 genes could be modelled
HD	GL945477.1: 1,592,030–1,592,212	undetermined	HD1	0	fewer than 2 genes could be modelled
PR	GL945478.1: 522,678–539,217	undetermined	pheromone_receptor, fungal_mating_type_pheromone	1	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
HD	GL945491.1: 329,751–329,885	undetermined	HD2	0	fewer than 2 genes could be modelled
PR	GL945492.1: 4,562–34,453	undetermined	pheromone_receptor, fungal_mating_type_pheromone	0	fewer than 2 genes could be modelled
HD	GL945482.1: 619,108–619,200	undetermined	HD1	0	fewer than 2 genes could be modelled
HD	GL945486.1: 1,083,087–1,083,326	undetermined	HD2	0	fewer than 2 genes could be modelled
HD	GL945486.1: 1,329,566–1,352,640	undetermined	HD1	0	fewer than 2 genes could be modelled
HD	GL945483.1: 85,456–85,572	undetermined	HD2	0	fewer than 2 genes could be modelled
HD	GL945483.1: 719,502–719,630	undetermined	HD2	0	fewer than 2 genes could be modelled
HD	GL945513.1: 5,399–10,253	undetermined	HD2	0	fewer than 2 genes could be modelled
HD	GL945495.1: 62,539–62,697	undetermined	HD1	0	fewer than 2 genes could be modelled
HD	GL945497.1: 182,086–182,313	undetermined	HD2	0	fewer than 2 genes could be modelled
HD	GL945613.1:128–217	undetermined	HD1	0	fewer than 2 genes could be modelled
PR	GL945488.1: 783,106–787,667	undetermined	pheromone_receptor, caax_precursor	2	too few of the family's genes found
PR	GL945491.1: 543,584–550,724	undetermined	pheromone_receptor, caax_precursor	1	too few of the family's genes found

Provenance

GENOME FILE

basidio_serpula_HD_PR.fna

ASSEMBLY

201 contigs · 46.1 Mb · N50 2.7 Mb

GENOME SHA-256

f6fabdee1e313be7d2be1ee5610
ae93ec85091955df52f79792f76
a72d7c1972

MATPREDICT

0.6.1

DATABASE

2b90ce832408b109 · 163 records

TAXONOMY

NCBI E-utilities

PARAMETERS

min_hits=2,
require_core_role=True,
max_polished_clusters_per_family=6,
polish_strong_identity=50.0

RUN STARTED

2026-10-08T04:22:53Z

RUN TIME

86 s

FILES

detection_report.yaml, detected_loci.gff3

Cite MATPredict as given in CITATION.cff at github.com/stajichlab/MATPredict. The reference records named in the gene tables are curated database entries; each record's metadata cites its source publication.

How to read this report

Idiomorph

The form of the mating-type locus a strain carries (for example MAT1-1 or MAT1-2, Plus or Minus). Basidiomycota HD and PR alleles are named after the reference they match.

High confidence

Every expected core MAT gene was found and modelled, with a conserved flanking gene where the family has one, on one contig.

Medium confidence

The core genes were found, but a gene could not be modelled, no conserved flank was found, or the locus is split (one step down).

Low confidence

A single isolated gene hit, or no core MAT gene.

Model status

Gene model: the reference protein was aligned to the genome to build the gene (exonerate, miniprot). Models differ: the two tools disagree on the structure; the alternate model is shown. Hit only: a search hit with no model, weaker evidence.

Identity

Percent amino-acid identity between the gene (or hit) and the closest reference protein.

Reference record

The curated database record whose protein matched best. Each record's metadata cites the publication or deposit it comes from.

LIMITS

- MATPredict reads an assembly; it is not a mating assay. Two idiomorphs in one assembly can come from a homothallic species, a mixed culture or a merged diploid assembly.
- Fragmented assemblies can split a locus or lose its flanking genes; an uncalled genome is weak evidence of absence.
- Lineages with no curated record are searched with the whole phylum's references; family labels there are less reliable.